Class-Based Modelling

Ramna Yoga Society Management System

Analysis stage	Confirmed result
Total nouns	250
Solution-space nouns	154
GC-passed nouns	50
SC-passed final classes	18
Approved CRC diagrams	17

Prepared from the approved editable class-based modelling source and CRC diagram files.

Document Structure

1. List of Total Nouns
2. List of Nouns in Solution Space
3. General Classifications (GC)
4. Selection Criteria (SC)
5. Associated Noun and Verb Identification
6. Attribute Selection
7. Method Identification
8. Finalizing Classes
9. Class Cards
10. CRC Diagrams

Modelling Rule

The total-noun list contains noun phrases explicitly extracted from the usage scenario. The solution-space list is a subset containing nouns represented or used by the proposed system. A solution-space noun passes General Classification when it satisfies at least three classifications or satisfies External Entity by itself. Only GC-passed nouns are evaluated by the Selection Criteria. A GC candidate passes SC when it satisfies at least three criteria or satisfies Essential Requirement. No separate merge stage is used; accepted source nouns are standardized into class names during finalization.

1. List of Total Nouns

The following 250 nouns and noun phrases are extracted from the usage scenario. They are listed before deciding whether the proposed system will represent or use them.

No.	Noun
1	Ramna Yoga Society Management System
2	System
3	Digital support
4	Membership
5	Attendance
6	Payment
7	Finance
8	Event
9	Notice
10	Yoga video
11	Equipment
12	Committee activity
13	Yoga knowledge assistance
14	Regular yoga session
15	Physical yoga session
16	Registered member
17	Authorized committee member
18	System role
19	General participant
20	Public information
21	Yoga activity
22	Regular yoga-session schedule
23	Schedule
24	Time
25	Location

No.	Noun
26	Public notice
27	Upcoming event
28	Public yoga video
29	General safety guidance
30	Account
31	Membership application
32	Membership applicant
33	Membership registration
34	Applicant
35	Digital membership form
36	Required personal information
37	Photograph
38	Application
39	Treasurer
40	Executive Committee
41	Information
42	Verification
43	Formal approval
44	Registration payment
45	Membership creation
46	BDT 500 registration fee
47	Supported payment facility
48	Accepted payment method
49	Payment received
50	Membership record
51	Unique membership number
52	QR code
53	Physical membership card
54	Member dashboard
55	Personal profile information
56	Membership status
57	Attendance history
58	Payment history
59	Donation history
60	Upcoming session

No.	Noun
61	Available yoga video
62	Payments and donations
63	Member
64	Monthly payment
65	Yearly payment
66	Donation
67	Applicable event payment
68	Successful payment
69	Corresponding member
70	Digital confirmation
71	Receipt
72	Event participation
73	Event registration
74	Advance registration
75	QR-based attendance
76	Attendance management
77	Selected yoga session
78	Selected event
79	Registered member's QR card
80	Member identity
81	Member status
82	Attendance record
83	Duplicate attendance
84	Unavailable QR card
85	Manual attendance
86	Member's membership number
87	Attendance summary
88	High attendance
89	Session management
90	Yoga-session date
91	Starting time
92	Instructor
93	Session status
94	Normal session time
95	One hour

No.	Noun
96	Fajr prayer
97	Committee administration
98	System permission
99	Official position
100	Committee membership
101	Administrative permission
102	Previous committee record
103	Membership and payment management
104	Membership applications
105	Applicant information
106	Membership records
107	Payments
108	Donations
109	Online-payment information
110	Overdue membership
111	Financial management
112	Income
113	Expense
114	Financial total
115	Balance information
116	Monthly financial report
117	Yearly financial report
118	President
119	Society responsibility structure
120	Financial record
121	Secretary
122	Organizational supervision
123	Membership summary
124	Financial summary
125	Event summary
126	Notice summary
127	Organizational summary
128	Important committee activity
129	Financial-recording responsibility
130	Report review

No.	Noun
131	Financial report
132	Attendance information
133	Membership information
134	Organizational report
135	Yoga content
136	Safety guidance
137	Yoga Instructor
138	Yoga-pose information
139	Instruction
140	Difficulty level
141	Possible risk
142	Age-related precaution
143	Instructor-approved guidance
144	Pose
145	Medical suitability
146	Particular person
147	Yoga video management
148	Approved yoga video
149	Video title
150	Video instructor
151	Video category
152	Video difficulty level
153	Safety information
154	Public video visibility
155	Members-only video visibility
156	User
157	Vajra
158	AI Yoga Knowledge Assistant
159	Member question
160	General visitor
161	Question
162	Yoga pose
163	Pose instruction
164	Precaution
165	Instructor information

No.	Noun
166	Yoga information
167	Approved instructor-provided content
168	Short conversational answer
169	Relevant video
170	Learning material
171	Safety-related question
172	Authorized yoga instructor
173	Medical condition
174	Specific person
175	Publicity Secretary
176	Notice and communication management
177	Official notice
178	Regular activity
179	Session change
180	Postponement
181	Cancellation
182	Announcement
183	User group
184	Previous notice
185	Media management
186	Activity post
187	Description
188	Activity photograph
189	Activity video
190	Activity date
191	Participation information
192	Media
193	Yoga-session media
194	Event media
195	Public media visibility
196	Members-only media visibility
197	Cultural Secretary
198	Event management
199	Cultural event
200	Special event

No.	Noun
201	Event date
202	Event time
203	Event location
204	Event description
205	Event instruction
206	Registration information
207	Participant information
208	Authorized user
209	Financial matter
210	Event publicity
211	Field and Equipment Secretary interaction
212	Field and Equipment Secretary
213	Equipment management
214	Equipment information
215	Quantity
216	Condition
217	Purchase information
218	Storage location
219	Equipment assignment
220	Regular-session equipment assignment
221	Event equipment assignment
222	Helping-Hand Management
223	Task
224	Helping hand
225	Yoga area
226	Equipment arrangement
227	Event support
228	Equipment return
229	Responsible committee member
230	Task progress
231	Helping-hand payment information
232	Automatic system interaction
233	Member engagement
234	QR attendance
235	Motivational message

No.	Noun
236	Attendance streak
237	Attendance milestone
238	Birthday
239	Membership anniversary
240	Birthday greeting
241	Membership-anniversary greeting
242	Special-day appearance
243	System appearance
244	Selected occasion
245	International Mother Language Day
246	Independence Day
247	International Yoga Day
248	Victory Day
249	Visual change
250	Regular system operation

Total extracted nouns: 250.

2. List of Nouns in Solution Space

The following 154 nouns are selected from the total-noun list because they are represented or used by the system as potential classes, roles, attributes, records, values, relationships, interface information, or method concepts. The 96 total nouns not included here are implicitly problem-space nouns; they are mentioned in the scenario but are not retained as system concepts.

No.	Solution-space noun
1	Account
2	Member
3	Yoga Instructor
4	Authorized committee member
5	President
6	Secretary
7	Treasurer
8	Publicity Secretary
9	Cultural Secretary
10	Field and Equipment Secretary
11	Committee activity
12	Regular yoga session
13	Event
14	Financial management
15	Supported payment facility
16	Activity post
17	Yoga content
18	AI Yoga Knowledge Assistant
19	General participant
20	Membership applicant
21	Applicant
22	Registered member
23	Authorized user
24	General visitor
25	Executive Committee
26	Membership
27	Membership application
28	Membership registration
29	Event registration
30	Event participation

No.	Solution-space noun
31	Attendance
32	Attendance record
33	Payment
34	Registration payment
35	Donation
36	Membership creation
37	Financial report
38	Notice
39	Yoga video
40	Equipment management
41	Equipment assignment
42	Committee membership
43	Committee administration
44	Attendance management
45	Session management
46	Event management
47	Member dashboard
48	Member engagement
49	Automatic system interaction
50	System appearance
51	System
52	System role
53	Public information
54	Schedule
55	Time
56	Location
57	Public notice
58	Upcoming event
59	General safety guidance
60	Digital membership form
61	Required personal information
62	Photograph
63	Application
64	Information
65	Verification

No.	Solution-space noun
66	BDT 500 registration fee
67	Accepted payment method
68	Payment received
69	Membership record
70	Unique membership number
71	QR code
72	Physical membership card
73	Personal profile information
74	Membership status
75	Attendance history
76	Payment history
77	Donation history
78	Upcoming session
79	Payments and donations
80	Monthly payment
81	Yearly payment
82	Applicable event payment
83	Successful payment
84	Digital confirmation
85	Receipt
86	Advance registration
87	System permission
88	Official position
89	Administrative permission
90	Previous committee record
91	QR-based attendance
92	Selected yoga session
93	Selected event
94	Registered member's QR card
95	Member identity
96	Member status
97	Duplicate attendance
98	Unavailable QR card
99	Manual attendance
100	Member's membership number

No.	Solution-space noun
101	Attendance summary
102	High attendance
103	Yoga-session date
104	Starting time
105	Instructor
106	Session status
107	Yoga activity
108	Regular yoga-session schedule
109	Attendance information
110	Membership information
111	Applicant information
112	Online-payment information
113	Overdue membership
114	Income
115	Expense
116	Financial total
117	Balance information
118	Monthly financial report
119	Yearly financial report
120	Financial record
121	Membership summary
122	Financial summary
123	Question
124	Yoga-pose information
125	Pose instruction
126	Difficulty level
127	Possible risk
128	Precaution
129	Age-related precaution
130	Instructor-approved guidance
131	Approved yoga video
132	Video title
133	Video category
134	Video difficulty level
135	Safety information

No.	Solution-space noun
136	Learning material
137	Official notice
138	Media management
139	Activity photograph
140	Activity video
141	Participation information
142	Event date
143	Event time
144	Event location
145	Event instruction
146	Registration information
147	Participant information
148	Event publicity
149	Equipment information
150	Quantity
151	Condition
152	Purchase information
153	Storage location
154	Equipment return

Solution-space result: 154 of 250 nouns are retained in the solution space.

3. General Classifications (GC)

All 154 solution-space nouns are considered. A noun passes GC when it satisfies at least three classifications, or when it satisfies External Entity by itself. The 50 passing nouns are shown individually below; the remaining 104 stay available as attributes, records, values, relationships, interface information, or method concepts.

3.1 Seven General Classifications

No.	General Classification	Description
1	External entities	People, organizations, devices, or external services that exchange information with the system.
2	Things	Physical or informational objects that the system stores, manages, or manipulates.
3	Events	Occurrences, transactions, or lifecycle changes that the system must recognize or record.
4	Roles	Capacities performed by people or software actors.
5	Organizational units	Groups or functional management units with coordinated responsibilities.
6	Places	Physical or logical locations relevant to system interaction.
7	Structures	Composite records, aggregates, hierarchies, or organized information structures.

3.2 Nouns Passing General Classification

No.	Solution-space noun	Accepted GC	Count	Result
1	Account	2, 4, 7	3	Pass
2	Member	1	1	Pass
3	Yoga Instructor	1	1	Pass
4	Authorized committee member	1	1	Pass
5	President	1	1	Pass
6	Secretary	1	1	Pass
7	Treasurer	1	1	Pass
8	Publicity Secretary	1	1	Pass
9	Cultural Secretary	1	1	Pass
10	Field and Equipment Secretary	1	1	Pass
11	Committee activity	2, 3, 7	3	Pass
12	Regular yoga session	2, 3, 7	3	Pass
13	Event	2, 3, 7	3	Pass
14	Financial management	2, 3, 5, 7	4	Pass
15	Supported payment facility	1, 2, 7	3	Pass
16	Activity post	2, 3, 7	3	Pass
17	Yoga content	2, 3, 7	3	Pass
18	AI Yoga Knowledge Assistant	1, 2, 4, 7	4	Pass
19	General participant	1	1	Pass
20	Membership applicant	1	1	Pass
21	Applicant	1	1	Pass
22	Registered member	1	1	Pass
23	Authorized user	1	1	Pass
24	General visitor	1	1	Pass
25	Executive Committee	1, 5, 7	3	Pass
26	Membership	2, 3, 7	3	Pass
27	Membership application	2, 3, 7	3	Pass
28	Membership registration	2, 3, 7	3	Pass
29	Event registration	2, 3, 7	3	Pass
30	Event participation	2, 3, 7	3	Pass
31	Attendance	2, 3, 7	3	Pass
32	Attendance record	2, 3, 7	3	Pass
33	Payment	2, 3, 7	3	Pass
34	Registration payment	2, 3, 7	3	Pass

No.	Solution-space noun	Accepted GC	Count	Result
35	Donation	2, 3, 7	3	Pass
36	Membership creation	2, 3, 7	3	Pass
37	Financial report	2, 3, 7	3	Pass
38	Notice	2, 3, 7	3	Pass
39	Yoga video	2, 3, 7	3	Pass
40	Equipment management	2, 3, 5, 7	4	Pass
41	Equipment assignment	2, 3, 7	3	Pass
42	Committee membership	2, 3, 4, 5, 7	5	Pass
43	Committee administration	2, 3, 4, 5, 7	5	Pass
44	Attendance management	2, 3, 5, 7	4	Pass
45	Session management	2, 3, 5, 7	4	Pass
46	Event management	2, 3, 5, 7	4	Pass
47	Member dashboard	2, 6, 7	3	Pass
48	Member engagement	2, 3, 7	3	Pass
49	Automatic system interaction	2, 3, 7	3	Pass
50	System appearance	2, 3, 7	3	Pass

GC result: 50 of the 154 solution-space nouns pass General Classification.

4. Selection Criteria (SC)

Only the 50 GC-passed nouns are evaluated. A candidate passes when it satisfies at least three criteria or satisfies Essential Requirement. Eighteen candidates pass and are standardized into the final class names; thirty-two fail as independent classes but may still contribute attributes, records, or behaviour to a selected class.

4.1 Six Selection Criteria

No.	Selection Criterion	Description
1	Retained Information	Information about the candidate must be remembered by the system.
2	Needed Services	The candidate provides identifiable services required by the system.
3	Multiple Attributes	The candidate needs several meaningful attributes.
4	Common Attributes	Instances or related subclasses share attributes.
5	Common Operations	Instances or related subclasses share operations.
6	Essential Requirement	An independent class is essential to satisfy the requirement.

4.2 Selection-Criteria Results

No.	GC-passed noun	Accepted SC	Points	Result	Final class
1	Account	1, 2, 3, 4, 5, 6	6	Pass	`UserAccount`
2	Member	1, 2, 3, 4, 5, 6	6	Pass	`Member`
3	Yoga Instructor	1, 2, 3, 5, 6	5	Pass	`YogaInstructor`
4	Authorized committee member	1, 2, 3, 4, 5, 6	6	Pass	`ExecutiveCommitteeMember`
5	President	1, 2, 3, 5, 6	5	Pass	`President`
6	Secretary	1, 2, 3, 5, 6	5	Pass	`Secretary`
7	Treasurer	1, 2, 3, 5, 6	5	Pass	`Treasurer`
8	Publicity Secretary	1, 2, 3, 5, 6	5	Pass	`PublicitySecretary`
9	Cultural Secretary	1, 2, 3, 5, 6	5	Pass	`CulturalSecretary`
10	Field and Equipment Secretary	1, 2, 3, 5, 6	5	Pass	`FieldEquipmentSecretary`
11	Committee activity	1, 2, 3, 4, 5, 6	6	Pass	`Activity`
12	Regular yoga session	1, 2, 3, 4, 5, 6	6	Pass	`YogaSession`
13	Event	1, 2, 3, 4, 5, 6	6	Pass	`Event`
14	Financial management	1, 2, 3, 4, 5, 6	6	Pass	`FinancialManagement`
15	Supported payment facility	2, 3, 5, 6	4	Pass	`PaymentGateway`
16	Activity post	1, 2, 3, 4, 5, 6	6	Pass	`MediaPost`
17	Yoga content	1, 2, 3, 4, 5, 6	6	Pass	`YogaContent`
18	AI Yoga Knowledge Assistant	1, 2, 3, 5, 6	5	Pass	`VajraAssistant`
19	General participant	2	1	Fail	-
20	Membership applicant	2, 3	2	Fail	-
21	Applicant	2, 3	2	Fail	-
22	Registered member	1, 3	2	Fail	-
23	Authorized user	2, 3	2	Fail	-
24	General visitor	2	1	Fail	-
25	Executive Committee	1, 3	2	Fail	-
26	Membership	1, 3	2	Fail	-
27	Membership application	1, 3	2	Fail	-
28	Membership registration	2, 3	2	Fail	-
29	Event registration	1, 3	2	Fail	-
30	Event participation	1, 3	2	Fail	-
31	Attendance	1, 3	2	Fail	-
32	Attendance record	1, 3	2	Fail	-
33	Payment	1, 3	2	Fail	-
34	Registration payment	1, 3	2	Fail	-

No.	GC-passed noun	Accepted SC	Points	Result	Final class
35	Donation	1, 3	2	Fail	-
36	Membership creation	2	1	Fail	-
37	Financial report	1, 3	2	Fail	-
38	Notice	1, 3	2	Fail	-
39	Yoga video	1, 3	2	Fail	-
40	Equipment management	2, 3	2	Fail	-
41	Equipment assignment	1, 3	2	Fail	-
42	Committee membership	1, 3	2	Fail	-
43	Committee administration	2, 3	2	Fail	-
44	Attendance management	2, 3	2	Fail	-
45	Session management	2, 3	2	Fail	-
46	Event management	2, 3	2	Fail	-
47	Member dashboard	2, 3	2	Fail	-
48	Member engagement	2, 3	2	Fail	-
49	Automatic system interaction	2, 5	2	Fail	-
50	System appearance	2, 3	2	Fail	-

SC result: 18 candidates pass and become the 18 final classes. Thirty-two candidates fail as independent classes.

5. Associated Noun and Verb Identification

Associated nouns indicate candidate attributes and relationships; associated verbs indicate candidate methods and responsibilities. The list remains domain-oriented and excludes framework, controller, and database implementation operations.

No.	Selected class	Associated nouns	Associated verbs
1	UserAccount	account identifier, username, password hash, account status, access permission, last login time	authenticate, log in, log out, change password
2	Member	member identifier, full name, contact information, photograph, date of birth, verification status, registration payment status, membership number, membership status, QR code, membership card, registration date, membership start date	register for membership, record verification, record registration payment, activate membership, generate membership number, generate QR code, issue card, update and view profile, view membership status, pay fee, donate, register for event, view attendance and financial histories, view activities, view media posts, view yoga content
3	YogaInstructor	instructor identifier, full name, contact information, expertise, yoga pose, instruction, difficulty, risk, precaution, video	create content, update content, add pose instruction, set difficulty, add risk, add precaution, attach demonstration video
4	ExecutiveCommitteeMember	committee member identifier, member, committee term, committee status, membership application, attendance, activity, summary, system permission	view membership application, verify member information, dismiss membership application, view activity, record QR attendance, record manual attendance, generate summary, view directory, access administration
5	President	president role, committee term, financial report, attendance summary, overdue member, committee activity, club operation	review report, review attendance, view overdue members, take action for overdue members, monitor committee, supervise operation
6	Secretary	secretary role, committee term, committee record, schedule, attendance summary, administrative activity	maintain record, coordinate schedule, review attendance, monitor administration
7	Treasurer	treasurer role, membership application, offline membership payment, donation, income, expense, balance, financial report, event financial details, overdue membership	view membership application, record offline membership payment, record donation, record income, record expense, view event financial details, generate report, identify overdue membership
8	PublicitySecretary	publicity role, notice, announcement, activity post, media, attachment, audience, visibility	create post, publish notice, publish activity media, update post, archive post

No.	Selected class	Associated nouns	Associated verbs
9	CulturalSecretary	cultural role, event, registration, registered member, event instruction, event financial details, event publicity request	view event, create event, update event, open and close registration, view registered members, submit event financial details, request event publicity
10	FieldEquipmentSecretary	field role, equipment catalogue, availability record, equipment assignment, return record	add equipment, update equipment, remove equipment, check availability, assign equipment, record return
11	Activity	activity identifier, title, date, start time, location, status, organizer, participant, attendance entry	schedule, reschedule, cancel, add organizer, record attendance, summarize attendance, list participant
12	YogaSession	session, instructor, duration, regular-session status, schedule	assign instructor, open session, close session, publish schedule
13	Event	event type, description, instruction, registration status, registered member, participation information	open registration, register member, remove registration, close registration, publish participation information
14	FinancialManagement	membership payment, donation, income entry, expense entry, balance, report, payment history, donation history	record payment, record donation, record income, record expense, calculate balance, reconcile transaction, generate report, retrieve history
15	PaymentGateway	gateway name, supported method, availability, payment request, transaction reference, transaction status, confirmation	process payment, verify transaction, confirm payment, obtain status
16	MediaPost	post identifier, post type, title, body, attachment, visibility, audience, publish date, related activity	create, update, publish, set audience, attach media, archive, link activity
17	YogaContent	content identifier, content type, title, instruction, difficulty, risk, precaution, video reference, visibility, approval status, author	create, update, add instruction, set difficulty, add risk, add precaution, attach video, publish, search
18	VajraAssistant	approved content source, question, interpreted topic, answer, suggestion, safety disclaimer	interpret question, search approved content, generate answer, suggest related content, attach safety disclaimer

6. Attribute Selection

Attributes are selected from the associated nouns and refined to avoid duplication. Subclasses inherit shared attributes from their superclasses. Collection-like attributes represent contained records without introducing unnecessary standalone classes at this level.

No.	Class	Selected attributes
1	UserAccount	accountId username passwordHash accountStatus accessPermissions lastLoginAt
2	Member	memberId fullName contactInformation photograph dateOfBirth verificationStatus registrationPaymentStatus membershipNumber membershipStatus qrCode cardInformation registrationDate membershipStartDate account
3	YogaInstructor	instructorId fullName contactInformation expertise account authoredContent
4	ExecutiveCommitteeMember	committeeMemberId member termStartDate termEndDate committeeStatus account assignedPermissions
5	President	inherits ExecutiveCommitteeMember attributes oversightNotes
6	Secretary	inherits ExecutiveCommitteeMember attributes committeeRecords coordinatedSchedules
7	Treasurer	inherits ExecutiveCommitteeMember attributes managedFinancialRecords
8	PublicitySecretary	inherits ExecutiveCommitteeMember attributes managedPosts
9	CulturalSecretary	inherits ExecutiveCommitteeMember attributes managedEvents
10	FieldEquipmentSecretary	inherits ExecutiveCommitteeMember attributes equipmentCatalogue availabilityRecords equipmentAssignments returnRecords

No.	Class	Selected attributes
11	Activity	activityId title activityDate startTime location status organizers participants attendanceEntries
12	YogaSession	inherits Activity attributes instructor duration regularSession
13	Event	inherits Activity attributes eventType description instructions registrationRequired registrationOpen registeredMembers participationInformation
14	FinancialManagement	membershipPayments donations incomeEntries expenseEntries currentBalance generatedReports
15	PaymentGateway	gatewayName supportedMethods availability transactionReferences lastGatewayResponse
16	MediaPost	postId postType title body attachments visibility targetAudience publishDate relatedActivity archiveStatus
17	YogaContent	contentId contentType title instructions difficultyLevel risks precautions videoReference visibility approvalStatus author createdAt updatedAt
18	VajraAssistant	approvedContentSource currentQuestion interpretedTopic generatedAnswer suggestions safetyDisclaimer

7. Method Identification

Methods are derived from associated verbs and assigned to the class that owns the required information. The method list describes domain behaviour; validation and persistence helpers are intentionally omitted.

No.	Class	Identified methods
1	UserAccount	authenticate() login() logout() changePassword()
2	Member	static registerForMembership() markInformationVerified() recordRegistrationPayment() activateMembership() generateMembershipNumber() generateQRCode() issueMembershipCard() updateProfile() payFee() donate() registerForEvent() viewProfile() viewMembershipStatus() viewAttendanceHistory() viewMediaPosts() viewPaymentHistory() viewDonationHistory() viewUpcomingActivities() viewYogaContent()
3	YogaInstructor	createYogaContent() updateYogaContent() addPoseInstructions() setDifficultyLevel() addRiskAndPrecaution() attachDemonstrationVideo() viewAssignedSessions()
4	ExecutiveCommitteeMember	viewMembershipApplication() verifyMemberInformation() dismissMembershipApplication() viewActivity() recordAttendanceByQR() recordAttendanceManually() generateAttendanceSummary() viewMemberDirectory() createActivity() updateActivity()
5	President	reviewFinancialReport() reviewAttendanceSummary() viewOverdueMembers() takeActionForOverdueMembers() monitorCommitteeActivities() superviseClubOperations()
6	Secretary	maintainCommitteeRecord() coordinateSchedule() reviewAttendanceSummary() monitorAdministrativeActivities()

No.	Class	Identified methods
7	Treasurer	recordMembershipPayment() recordDonation() recordIncome() recordExpense() viewEventFinancialDetails() generateFinancialReport() identifyOverdueMemberships()
8	PublicitySecretary	createMediaPost() publishNotice() publishActivityMedia() updatePost() archivePost() setTargetAudience()
9	CulturalSecretary	viewEvents() createEvent() updateEvent() openEventRegistration() closeEventRegistration() viewRegisteredMembers() submitEventFinancialDetails() requestEventPublicity()
10	FieldEquipmentSecretary	addEquipment() updateEquipment() removeEquipment() checkAvailability() assignEquipment() recordReturn()
11	Activity	schedule() reschedule() cancel() addOrganizer() recordAttendance() generateAttendanceSummary() listParticipants()
12	YogaSession	assignInstructor() openSession() closeSession() publishSessionSchedule()
13	Event	openRegistration() registerMember() removeRegistration() closeRegistration() publishParticipationInformation() listRegisteredMembers()
14	FinancialManagement	recordPayment() recordDonation() recordIncome() recordExpense() calculateBalance() reconcileTransaction() generateFinancialReport() getPaymentHistory() getDonationHistory() identifyOverdueMemberships()
15	PaymentGateway	processPayment() verifyTransaction() confirmPayment() getTransactionStatus() mapGatewayResponse()

No.	Class	Identified methods
16	MediaPost	create() update() publish() setAudience() attachMedia() archive() linkActivity()
17	YogaContent	create() update() addInstructions() setDifficulty() addRisk() addPrecaution() attachVideo() publish() search()
18	VajraAssistant	interpretQuestion() searchApprovedContent() generateAnswer() suggestRelatedContent() attachSafetyDisclaimer() clearConversationContext()

8. Finalizing Classes

The selected classes are finalized by checking responsibility placement, inheritance opportunities, and unnecessary duplication.

1. Authentication separation. Account credentials, status, and permission behaviour are centralized in `UserAccount`. `Member`, `YogaInstructor`, and `ExecutiveCommitteeMember` collaborate with it instead of duplicating authentication data.
2. Membership lifecycle. `Applicant`, `MembershipRegistration`, and `MembershipApplication` are not retained as separate classes. `Member` represents both pending and active membership states. The public `Member.registerForMembership(...)` operation creates the pending member record without requiring an account or login. The model has no separate submission, correction, or application-status-viewing operation.
3. Verification and activation rule. An `ExecutiveCommitteeMember` views the pending membership application and either verifies its information or dismisses it. Incorrect information is dismissed rather than returned for correction. Registration payment occurs only after verification. `PaymentGateway` confirms online payment automatically, while the Treasurer records an accepted offline payment. Once verification and payment are both recorded, the system invokes `activateMembership()`, `generateMembershipNumber()`, `generateQRCode()`, and `issueMembershipCard()` on the pending `Member` record. There is no separate formal approval operation.
4. Committee hierarchy. `ExecutiveCommitteeMember` is a concrete superclass representing general executive members. `President`, `Secretary`, `Treasurer`, `PublicitySecretary`, `CulturalSecretary`, and `FieldEquipmentSecretary` specialize it because they have distinct permissions and methods.
5. Activity hierarchy. `Activity` is abstract and contains shared schedule, organizer, participant, and attendance behaviour. `YogaSession` and `Event` are its concrete subclasses.
6. Attendance placement. `Attendance` and `AttendanceRecord` are absorbed into `Activity` as `attendanceEntries` and attendance operations. A separate Attendance class would duplicate the activity context.
7. Event participation. `EventParticipation` and `EventRegistration` are absorbed into `Event` as `registeredMembers`, `participationInformation`, and registration methods.
8. Financial consolidation. `FinancialTransaction`, payment, donation, income, expense, and reports are maintained by `FinancialManagement`. `PaymentGateway` remains separate because it represents the external payment boundary.
9. Communication consolidation. Notice, announcement, activity post, and event media share publication attributes and operations and are merged into `MediaPost`.
10. Yoga knowledge consolidation. Pose information, instructions, risks, precautions, and videos are consolidated in `YogaContent`. `VajraAssistant` remains separate because it provides conversational retrieval and safety guidance.
11. Equipment responsibility. Equipment catalogue, availability, assignment, and return information are owned directly by `FieldEquipmentSecretary`; a separate inventory-management class would duplicate that role. Operational tasks, task tracking, and `HelpingHand` are not retained in this high-level model.
12. Presentation and automation. `MemberDashboard`, `SystemAppearance`, and general `SystemAutomation` are boundary or presentation behaviours rather than independent domain classes.

13. Role-specific workflow. The `CulturalSecretary` manages events, submits event financial details to the `Treasurer`, and requests event publicity. The `Treasurer` records the resulting financial entries, while the `President` reviews overdue members and takes organizational action when required.

8.1 Final Class List

1. `UserAccount`
2. `Member`
3. `YogaInstructor`
4. `ExecutiveCommitteeMember`
5. `President`
6. `Secretary`
7. `Treasurer`
8. `PublicitySecretary`
9. `CulturalSecretary`
10. `FieldEquipmentSecretary`
11. `Activity`
12. `YogaSession`
13. `Event`
14. `FinancialManagement`
15. `PaymentGateway`
16. `MediaPost`
17. `YogaContent`
18. `VajraAssistant`

Final class count: 18.

9. Class Cards

Each card records the selected attributes and methods, followed by the class responsibilities and principal collaborators.

9.1 UserAccount

Attributes

- accountId
- username
- passwordHash
- accountStatus
- accessPermissions
- lastLoginAt

Methods

- authenticate()
- login()
- logout()
- changePassword()

Responsibility-Collaborator Mapping

Responsibility	Corresponding Collaborator(s)
Authentication	Member, YogaInstructor, ExecutiveCommitteeMember
Account and session management	Member, YogaInstructor, ExecutiveCommitteeMember
Role-based access control	Member, YogaInstructor, ExecutiveCommitteeMember

9.2 Member

Attributes

- memberId
- fullName
- contactInformation
- photograph
- dateOfBirth
- verificationStatus
- registrationPaymentStatus
- membershipNumber
- membershipStatus
- qrCode
- cardInformation
- registrationDate
- membershipStartDate
- account

Methods

- static registerForMembership()
- markInformationVerified()
- recordRegistrationPayment()
- activateMembership()
- generateMembershipNumber()
- generateQRCode()
- issueMembershipCard()
- updateProfile()
- payFee()
- donate()
- registerForEvent()
- viewProfile()
- viewMembershipStatus()
- viewAttendanceHistory()
- viewMediaPosts()
- viewPaymentHistory()
- viewDonationHistory()
- viewUpcomingActivities()

- viewYogaContent()

Responsibility-Collaborator Mapping

Responsibility	Corresponding Collaborator(s)
Authentication	UserAccount
Membership verification and activation	ExecutiveCommitteeMember, Treasurer
Member payment	FinancialManagement, PaymentGateway, Treasurer
Event registration	Event
Activity and attendance	Activity, ExecutiveCommitteeMember
Media-post viewing	MediaPost
Yoga-content access	YogaContent

9.3 YogaInstructor

Attributes

- instructorId
- fullName
- contactInformation
- expertise
- account
- authoredContent

Methods

- createYogaContent()
- updateYogaContent()
- addPoseInstructions()
- setDifficultyLevel()
- addRiskAndPrecaution()
- attachDemonstrationVideo()
- viewAssignedSessions()

Responsibility-Collaborator Mapping

Responsibility	Corresponding Collaborator(s)
Authentication	UserAccount
Yoga-session instruction and assignment	YogaSession
Yoga-content preparation and maintenance	YogaContent

9.4 ExecutiveCommitteeMember

Attributes

- committeeMemberId
- member
- termStartDate
- termEndDate
- committeeStatus
- account
- assignedPermissions

Methods

- viewMembershipApplication()
- verifyMemberInformation()
- dismissMembershipApplication()
- viewActivity()
- recordAttendanceByQR()
- recordAttendanceManually()
- generateAttendanceSummary()
- viewMemberDirectory()
- createActivity()
- updateActivity()

Responsibility-Collaborator Mapping

Responsibility	Corresponding Collaborator(s)
Authentication	UserAccount
Membership-application management	Member
Activity management	Activity
Activity and attendance	Activity, Member

9.5 President

Attributes

- inherits ExecutiveCommitteeMember attributes
- oversightNotes

Methods

- reviewFinancialReport()
- reviewAttendanceSummary()
- viewOverdueMembers()
- takeActionForOverdueMembers()
- monitorCommitteeActivities()
- superviseClubOperations()

Responsibility-Collaborator Mapping

Responsibility	Corresponding Collaborator(s)
Authentication	UserAccount
Committee and club supervision	ExecutiveCommitteeMember, Secretary
Activity and attendance oversight	Activity, ExecutiveCommitteeMember, Secretary
Financial oversight	FinancialManagement, Treasurer
Overdue-membership oversight and action	Member, Treasurer

9.6 Secretary

Attributes

- inherits ExecutiveCommitteeMember attributes
- committeeRecords
- coordinatedSchedules

Methods

- maintainCommitteeRecord()
- coordinateSchedule()
- reviewAttendanceSummary()
- monitorAdministrativeActivities()

Responsibility-Collaborator Mapping

Responsibility	Corresponding Collaborator(s)
Authentication	UserAccount
Committee-record management	ExecutiveCommitteeMember
Activity-schedule coordination	Activity, ExecutiveCommitteeMember
Activity and attendance	Activity, ExecutiveCommitteeMember
Administrative monitoring and coordination	President, ExecutiveCommitteeMember

9.7 Treasurer

Attributes

- inherits ExecutiveCommitteeMember attributes
- managedFinancialRecords

Methods

- recordMembershipPayment()
- recordDonation()
- recordIncome()
- recordExpense()
- viewEventFinancialDetails()
- generateFinancialReport()
- identifyOverdueMemberships()

Responsibility-Collaborator Mapping

Responsibility	Corresponding Collaborator(s)
Authentication	UserAccount
Membership-application access	ExecutiveCommitteeMember, Member
Membership-payment recording and confirmation	Member, FinancialManagement
Financial-transaction recording	Member, FinancialManagement
Event-finance review	CulturalSecretary, FinancialManagement
Financial reporting	FinancialManagement, President, Secretary
Overdue-membership identification	Member, FinancialManagement, President

9.8 PublicitySecretary

Attributes

- inherits ExecutiveCommitteeMember attributes
- managedPosts

Methods

- createMediaPost()
- publishNotice()
- publishActivityMedia()
- updatePost()
- archivePost()
- setTargetAudience()

Responsibility-Collaborator Mapping

Responsibility	Corresponding Collaborator(s)
Authentication	UserAccount
Media-post management	MediaPost
Official-notice publication	ExecutiveCommitteeMember, MediaPost
Activity publicity and media	Activity, MediaPost
Event publicity	Event, CulturalSecretary, MediaPost

9.9 CulturalSecretary

Attributes

- inherits ExecutiveCommitteeMember attributes
- managedEvents

Methods

- viewEvents()
- createEvent()
- updateEvent()
- openEventRegistration()
- closeEventRegistration()
- viewRegisteredMembers()
- submitEventFinancialDetails()
- requestEventPublicity()

Responsibility-Collaborator Mapping

Responsibility	Corresponding Collaborator(s)
Authentication	UserAccount
Committee administration	ExecutiveCommitteeMember
Event management	Event
Event registration and participant management	Event, Member
Event-finance submission	Event, Treasurer
Event-publicity coordination	Event, PublicitySecretary

9.10 FieldEquipmentSecretary

Attributes

- inherits ExecutiveCommitteeMember attributes
- equipmentCatalogue
- availabilityRecords
- equipmentAssignments
- returnRecords

Methods

- addEquipment()
- updateEquipment()
- removeEquipment()
- checkAvailability()
- assignEquipment()
- recordReturn()

Responsibility-Collaborator Mapping

Responsibility	Corresponding Collaborator(s)
Authentication	UserAccount
Committee administration	ExecutiveCommitteeMember
Equipment management for activities	Activity
Equipment assignment and return	Activity

9.11 Activity

Attributes

- activityId
- title
- activityDate
- startTime
- location
- status
- organizers
- participants
- attendanceEntries

Methods

- schedule()
- reschedule()
- cancel()
- addOrganizer()
- recordAttendance()
- generateAttendanceSummary()
- listParticipants()

Responsibility-Collaborator Mapping

Responsibility	Corresponding Collaborator(s)
Activity creation and scheduling	ExecutiveCommitteeMember
Activity-lifecycle management	ExecutiveCommitteeMember
Organizer management	ExecutiveCommitteeMember
Activity and attendance	Member, ExecutiveCommitteeMember
Activity-information publication	MediaPost

9.12 YogaSession

Attributes

- inherits Activity attributes
- instructor
- duration
- regularSession

Methods

- assignInstructor()
- openSession()
- closeSession()
- publishSessionSchedule()

Responsibility-Collaborator Mapping

Responsibility	Corresponding Collaborator(s)
Yoga-session preparation and scheduling	Activity, ExecutiveCommitteeMember
Instructor assignment	YogaInstructor, ExecutiveCommitteeMember
Session participation and attendance	Member, Activity, ExecutiveCommitteeMember
Session-schedule publication	Member, ExecutiveCommitteeMember

9.13 Event

Attributes

- inherits Activity attributes
- eventType
- description
- instructions
- registrationRequired
- registrationOpen
- registeredMembers
- participationInformation

Methods

- openRegistration()
- registerMember()
- removeRegistration()
- closeRegistration()
- publishParticipationInformation()
- listRegisteredMembers()

Responsibility-Collaborator Mapping

Responsibility	Corresponding Collaborator(s)
Event management	CulturalSecretary, Activity
Event registration	Member, CulturalSecretary
Participation and attendance management	Member, ExecutiveCommitteeMember, Activity
Event publicity	MediaPost, PublicitySecretary, CulturalSecretary
Event-finance information	FinancialManagement, CulturalSecretary, Treasurer

9.14 FinancialManagement

Attributes

- membershipPayments
- donations
- incomeEntries
- expenseEntries
- currentBalance
- generatedReports

Methods

- recordPayment()
- recordDonation()
- recordIncome()
- recordExpense()
- calculateBalance()
- reconcileTransaction()
- generateFinancialReport()
- getPaymentHistory()
- getDonationHistory()
- identifyOverdueMemberships()

Responsibility-Collaborator Mapping

Responsibility	Corresponding Collaborator(s)
Member-payment management	Member, Treasurer, PaymentGateway
Donation management	Member, Treasurer, PaymentGateway
Income-and-expense management	Treasurer
Balance calculation and transaction reconciliation	Treasurer, PaymentGateway
Payment and donation history	Member, Treasurer
Financial reporting	Treasurer, President, Secretary
Overdue-membership identification	Member, Treasurer, President
Event-finance management	Event, Treasurer

9.15 PaymentGateway

Attributes

- gatewayName
- supportedMethods
- availability
- transactionReferences
- lastGatewayResponse

Methods

- processPayment()
- verifyTransaction()
- confirmPayment()
- getTransactionStatus()
- mapGatewayResponse()

Responsibility-Collaborator Mapping

Responsibility	Corresponding Collaborator(s)
Online-payment processing	Member, FinancialManagement
Transaction-status checking	FinancialManagement
Transaction verification	Member, FinancialManagement
Payment confirmation	Member, FinancialManagement
Gateway-response adaptation	FinancialManagement

9.16 MediaPost

Attributes

- postId
- postType
- title
- body
- attachments
- visibility
- targetAudience
- publishDate
- relatedActivity
- archiveStatus

Methods

- create()
- update()
- publish()
- setAudience()
- attachMedia()
- archive()
- linkActivity()

Responsibility-Collaborator Mapping

Responsibility	Corresponding Collaborator(s)
Media-post management	PublicitySecretary
Activity-media linking	Activity, PublicitySecretary
Media-post viewing	Member
Post archiving	PublicitySecretary

9.17 YogaContent

Attributes

- contentId
- contentType
- title
- instructions
- difficultyLevel
- risks
- precautions
- videoReference
- visibility
- approvalStatus
- author
- createdAt
- updatedAt

Methods

- create()
- update()
- addInstructions()
- setDifficulty()
- addRisk()
- addPrecaution()
- attachVideo()
- publish()
- search()

Responsibility-Collaborator Mapping

Responsibility	Corresponding Collaborator(s)
Yoga-content preparation and maintenance	YogaInstructor
Yoga-instruction and safety-information management	YogaInstructor
Approved-content publication	YogaInstructor
Approved-content search	VajraAssistant
Yoga-content access	Member
Question-answering content supply	VajraAssistant

9.18 VajraAssistant

Attributes

- approvedContentSource
- currentQuestion
- interpretedTopic
- generatedAnswer
- suggestions
- safetyDisclaimer

Methods

- interpretQuestion()
- searchApprovedContent()
- generateAnswer()
- suggestRelatedContent()
- attachSafetyDisclaimer()
- clearConversationContext()

Responsibility-Collaborator Mapping

Responsibility	Corresponding Collaborator(s)
Member-question interpretation	Member
Approved-content search	YogaContent
Answer generation	Member, YogaContent
Related-content recommendation	Member, YogaContent
Safety-guidance presentation	Member, YogaContent, YogaInstructor

10. CRC Diagrams

Each diagram is reproduced from the approved PlantUML source in the CRC diagrams folder. Its classes, connections, layout intent, and collaborators are preserved; only relationship wording is aligned with the finalized workflows.

Ref: Table 10

Name: UserAccount

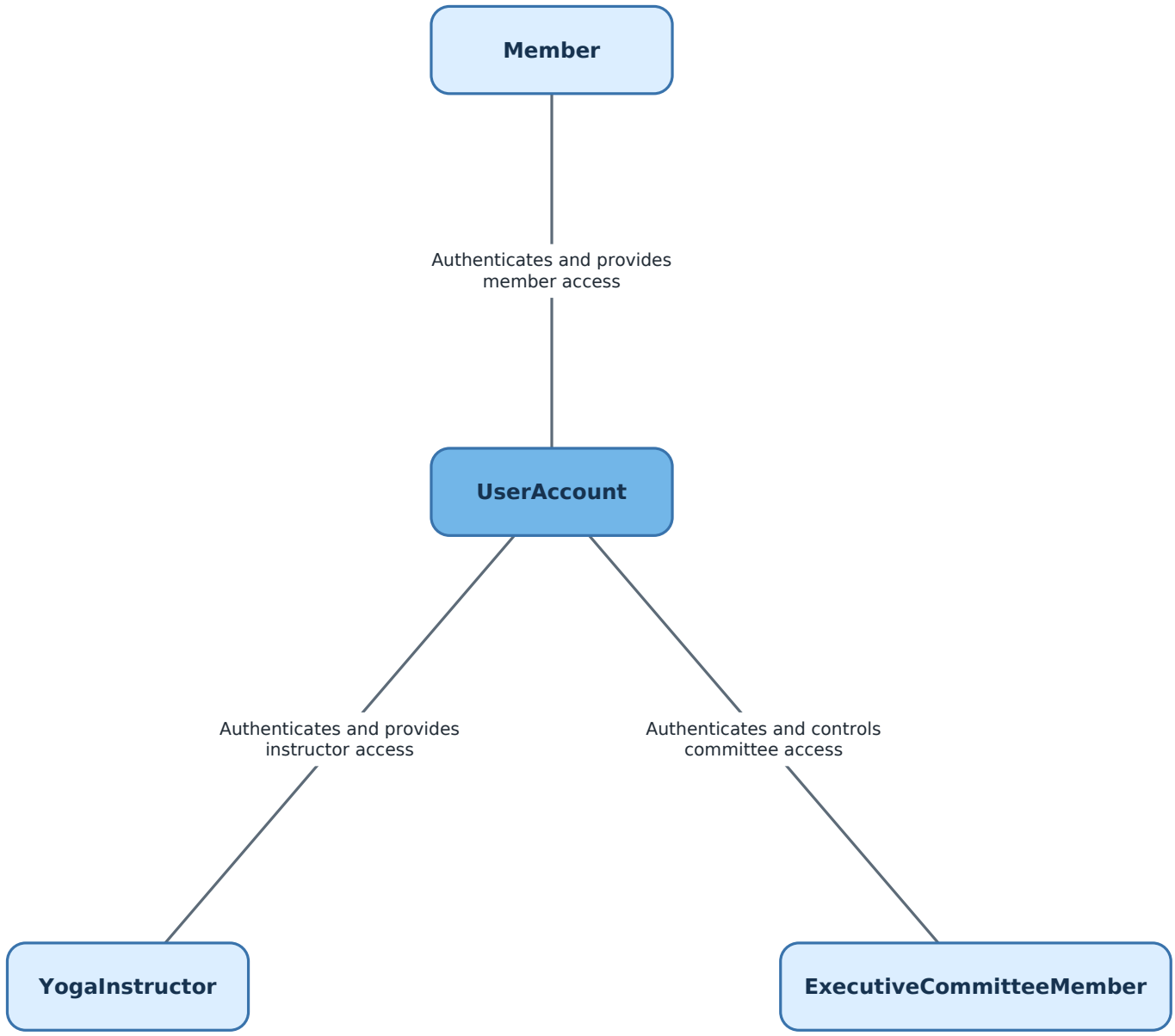


Fig crc.1: UserAccount

Ref: Table 11

Name: Member

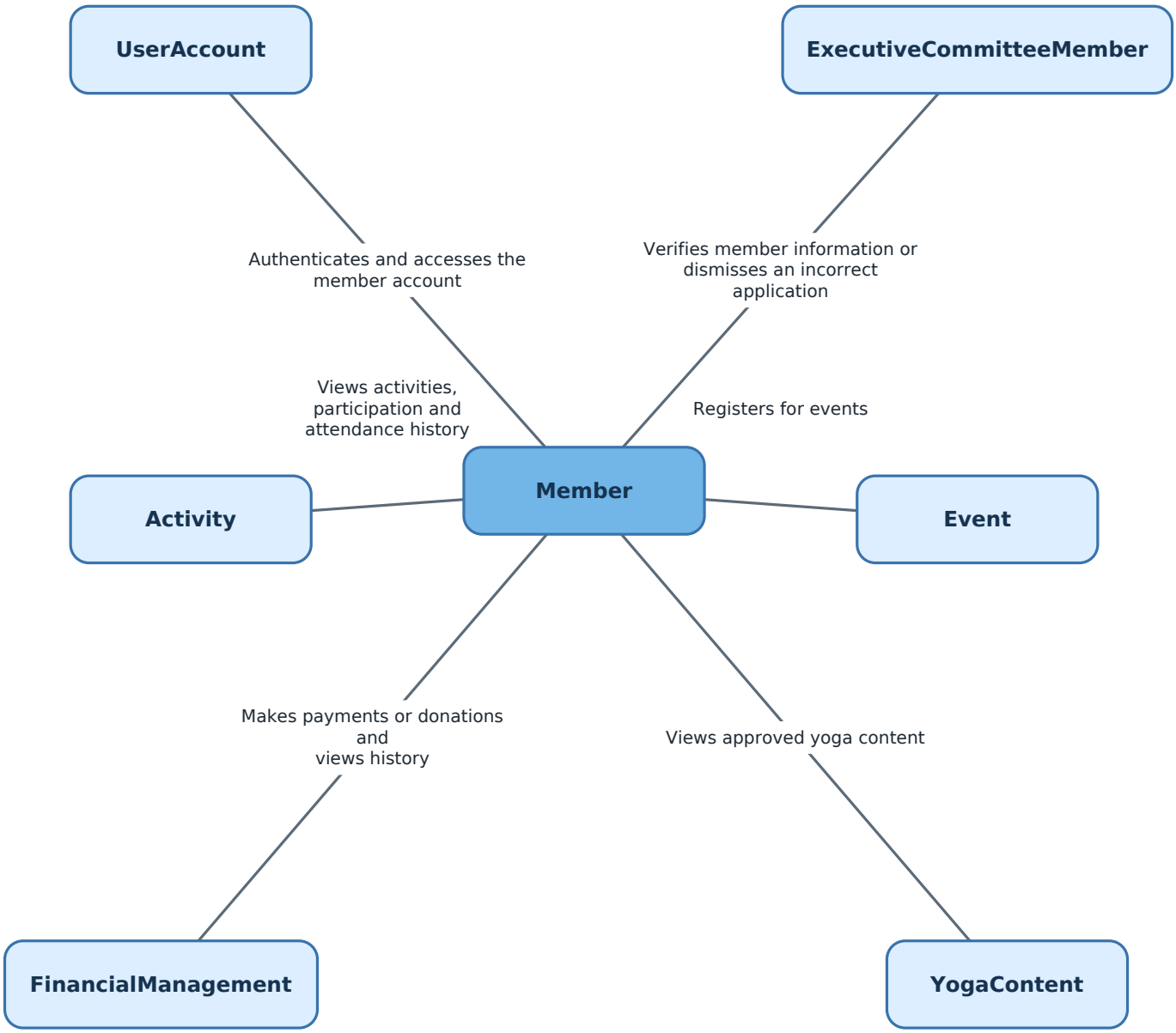


Fig crc.2: Member

Ref: Table 12

Name: YogaInstructor

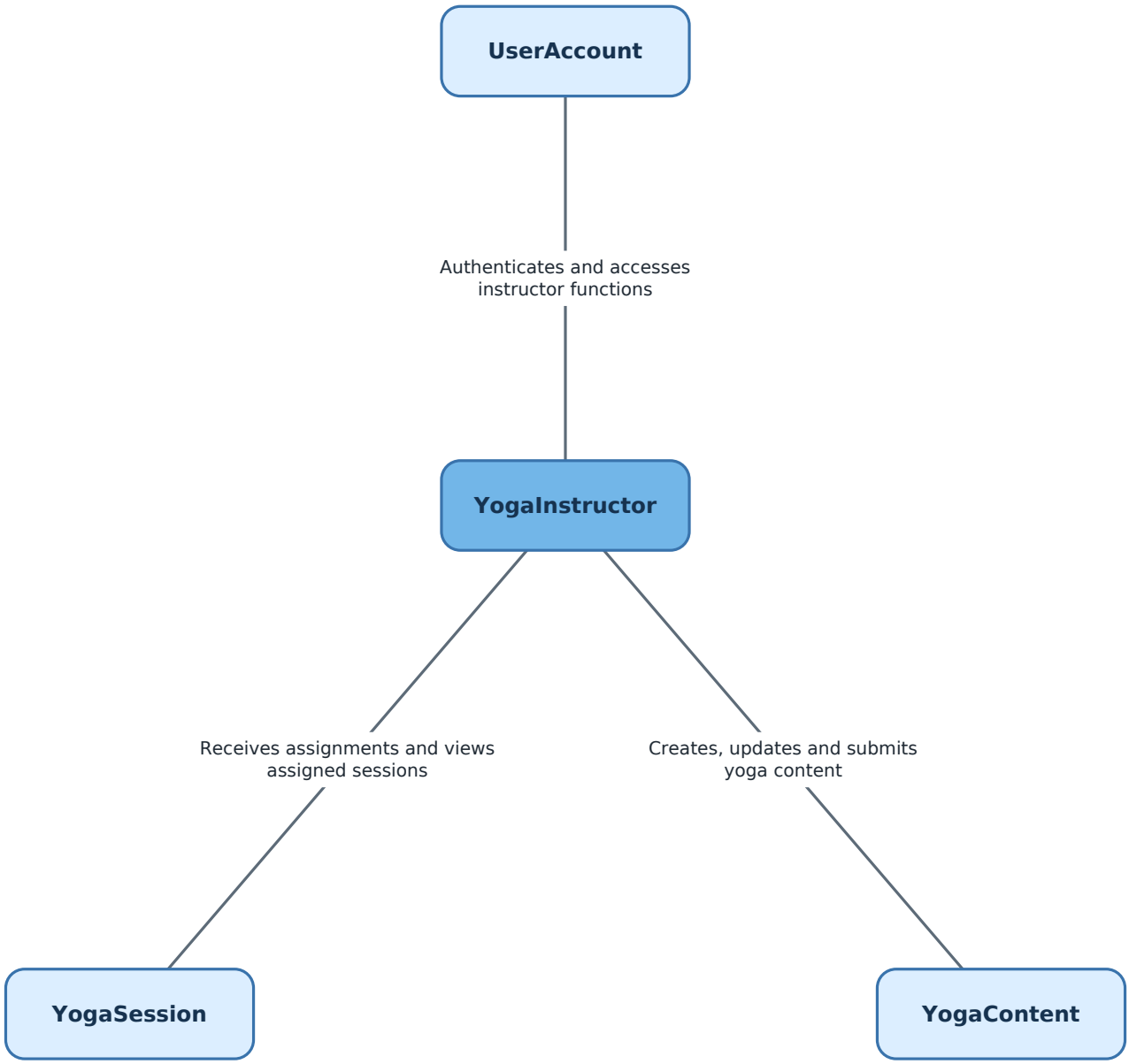


Fig crc.3: YogaInstructor

Ref: Table 13

Name: ExecutiveCommitteeMember

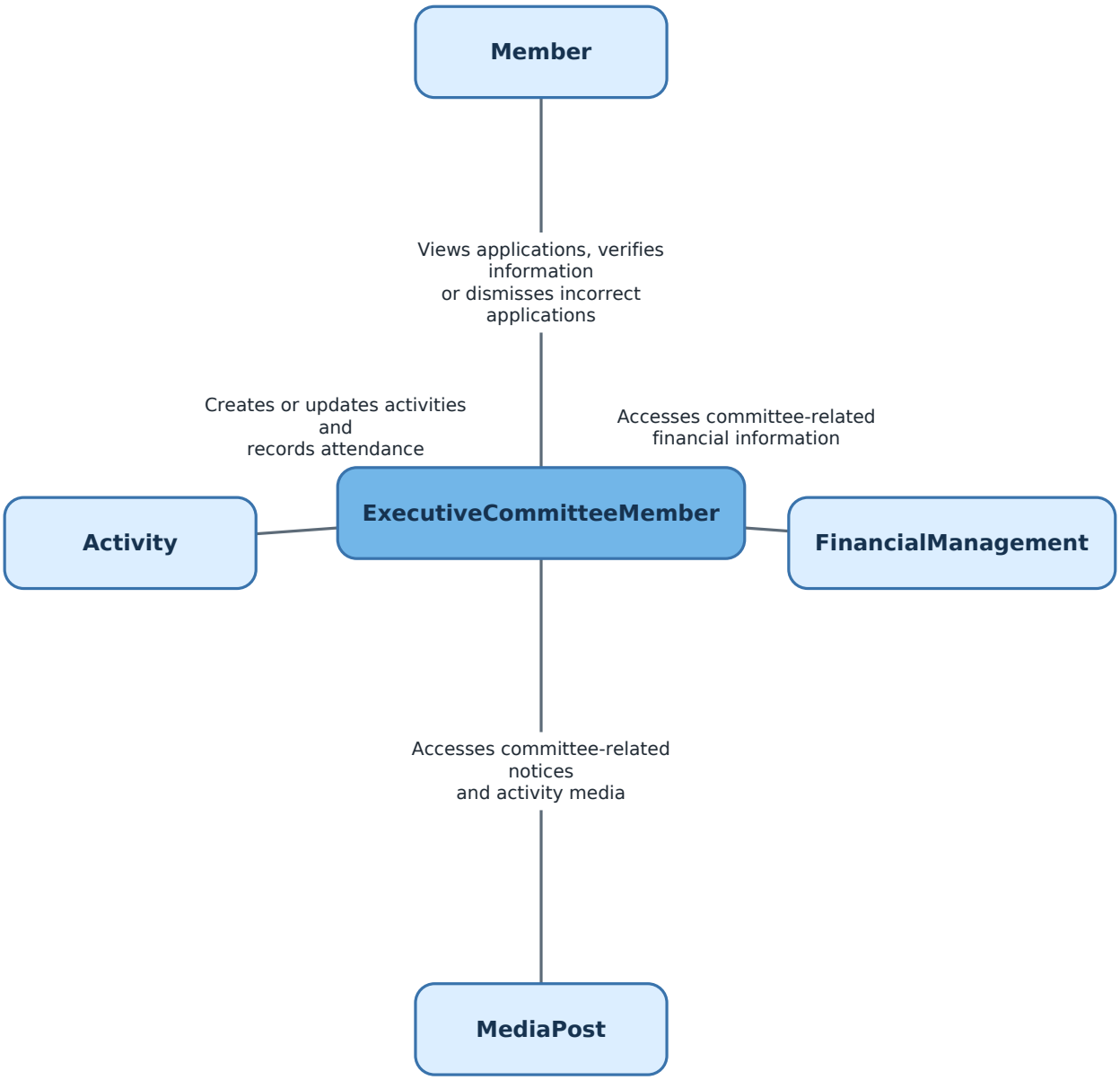


Fig crc.4: ExecutiveCommitteeMember

Ref: Table 14

Name: President

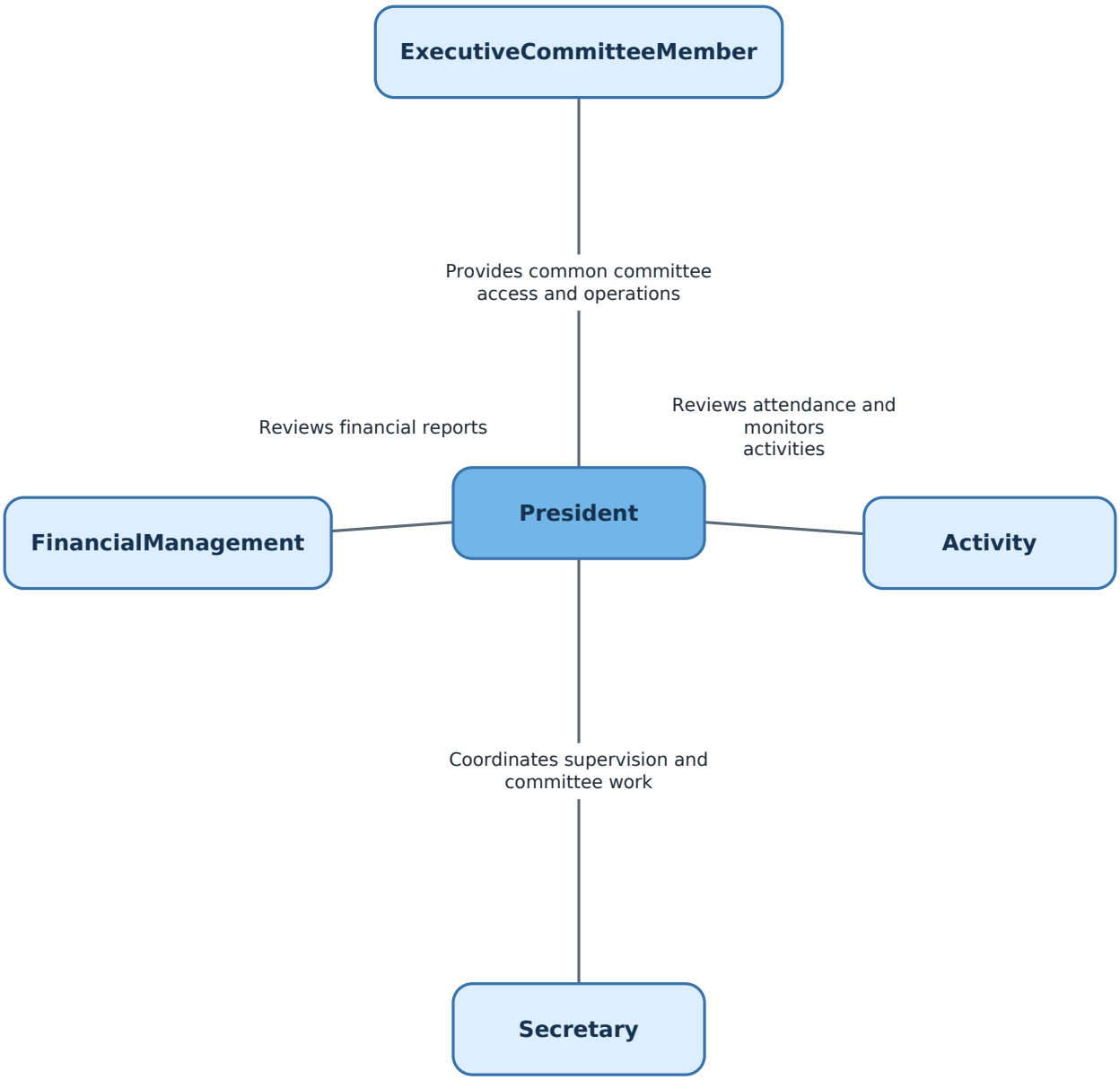


Fig crc.5: President

Ref: Table 15

Name: Secretary

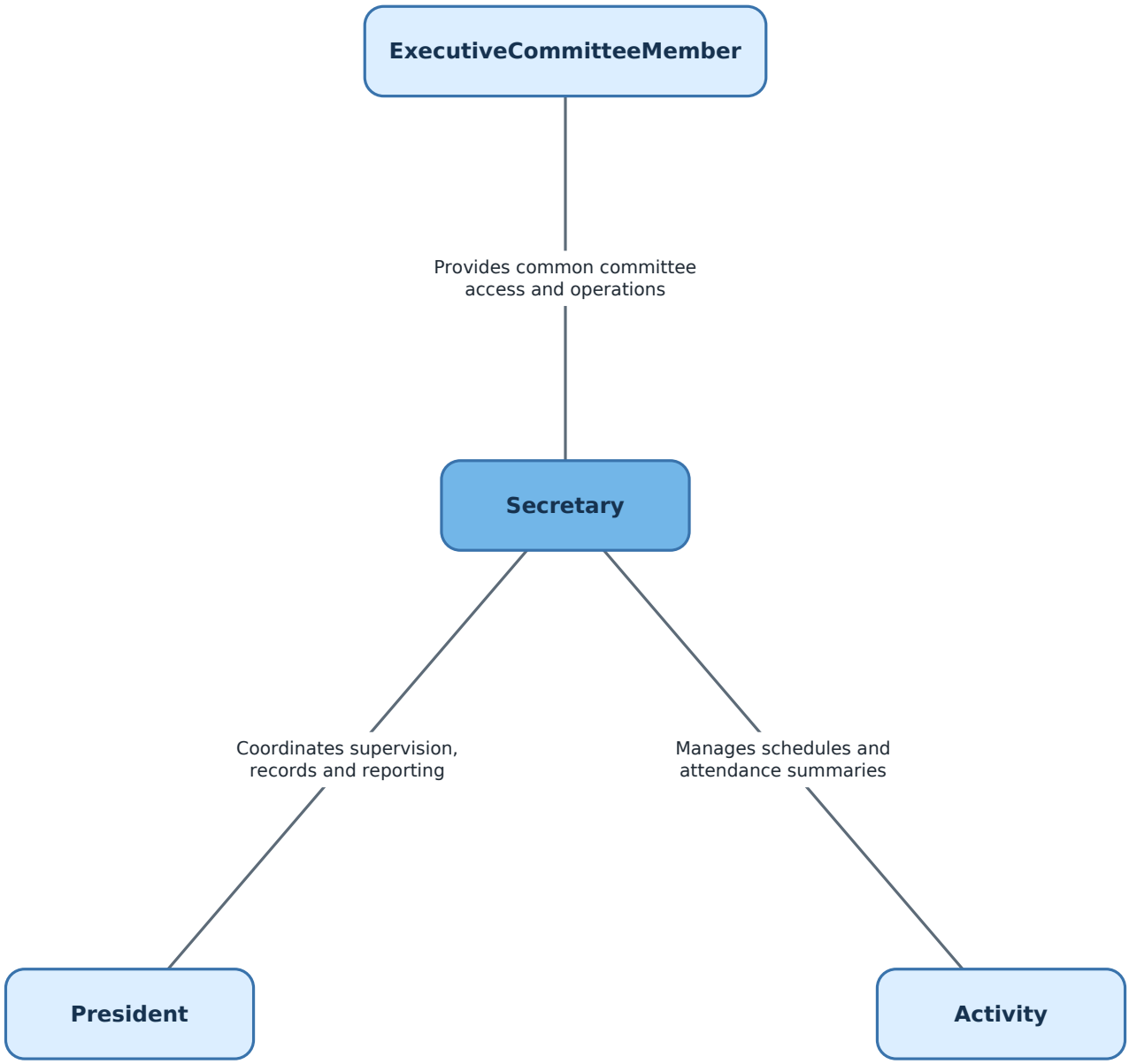


Fig crc.6: Secretary

Ref: Table 16

Name: Treasurer

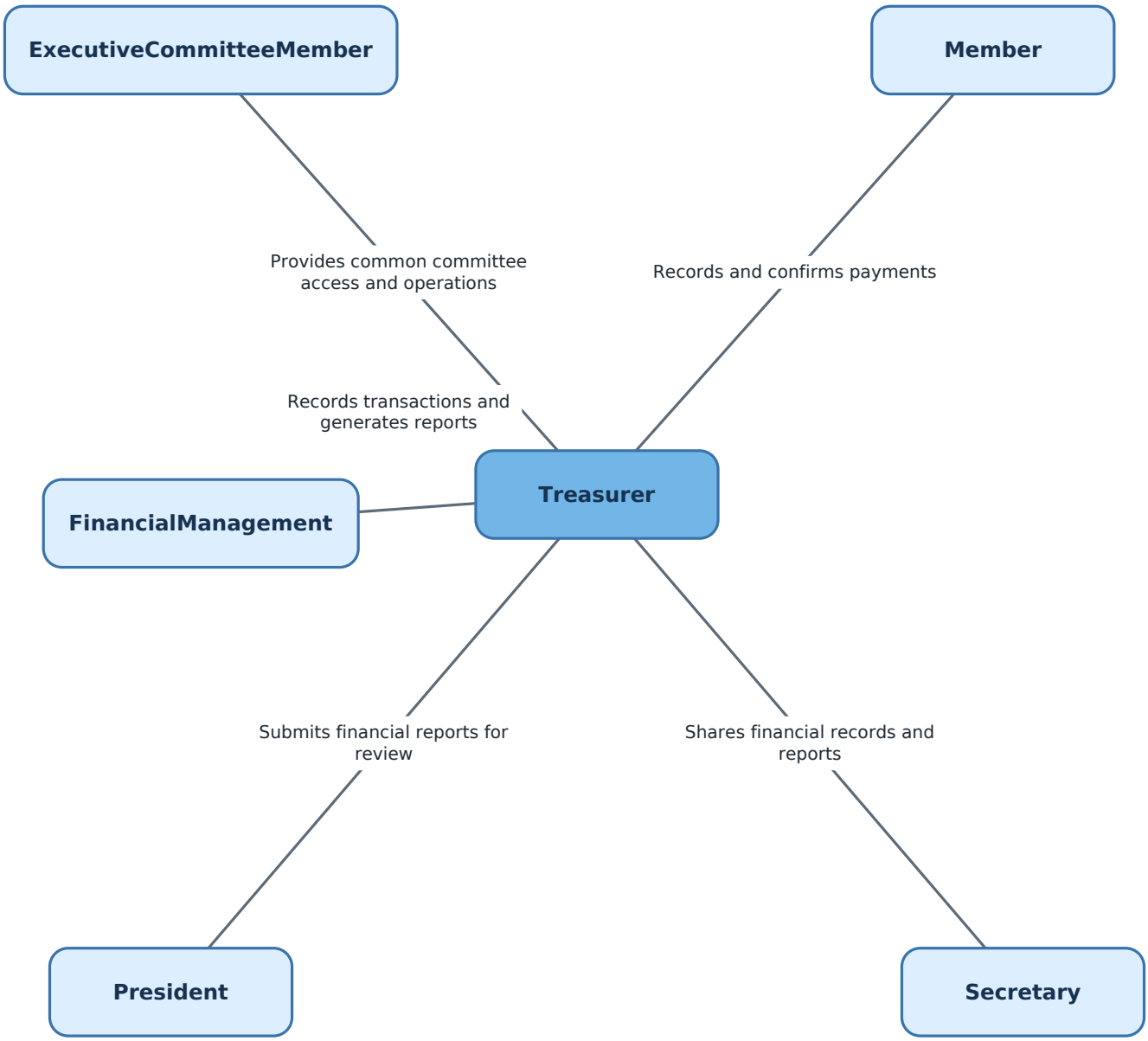


Fig crc.7: Treasurer

Ref: Table 17

Name: PublicitySecretary

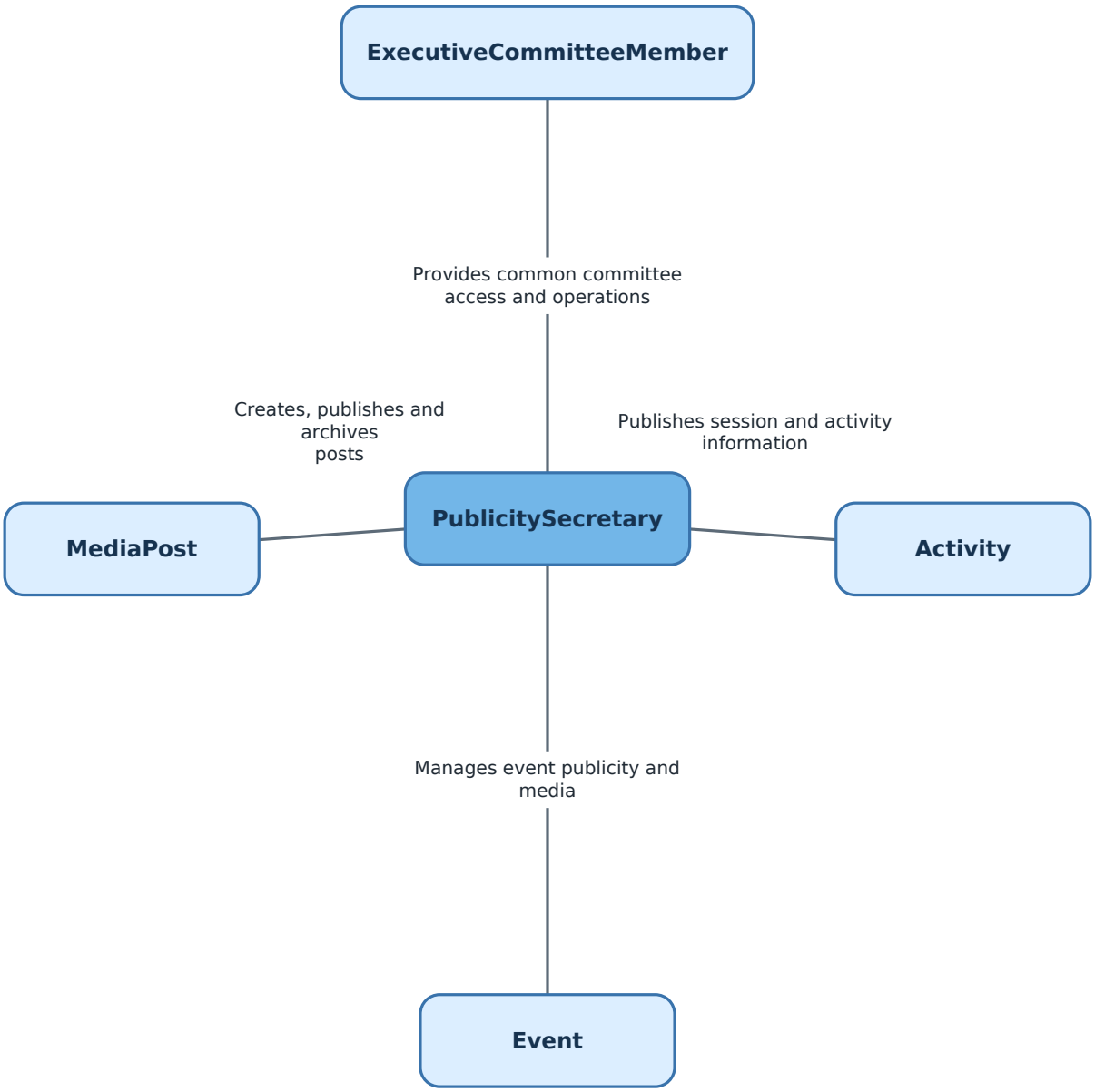


Fig crc.8: PublicitySecretary

Ref: Table 18

Name: CulturalSecretary

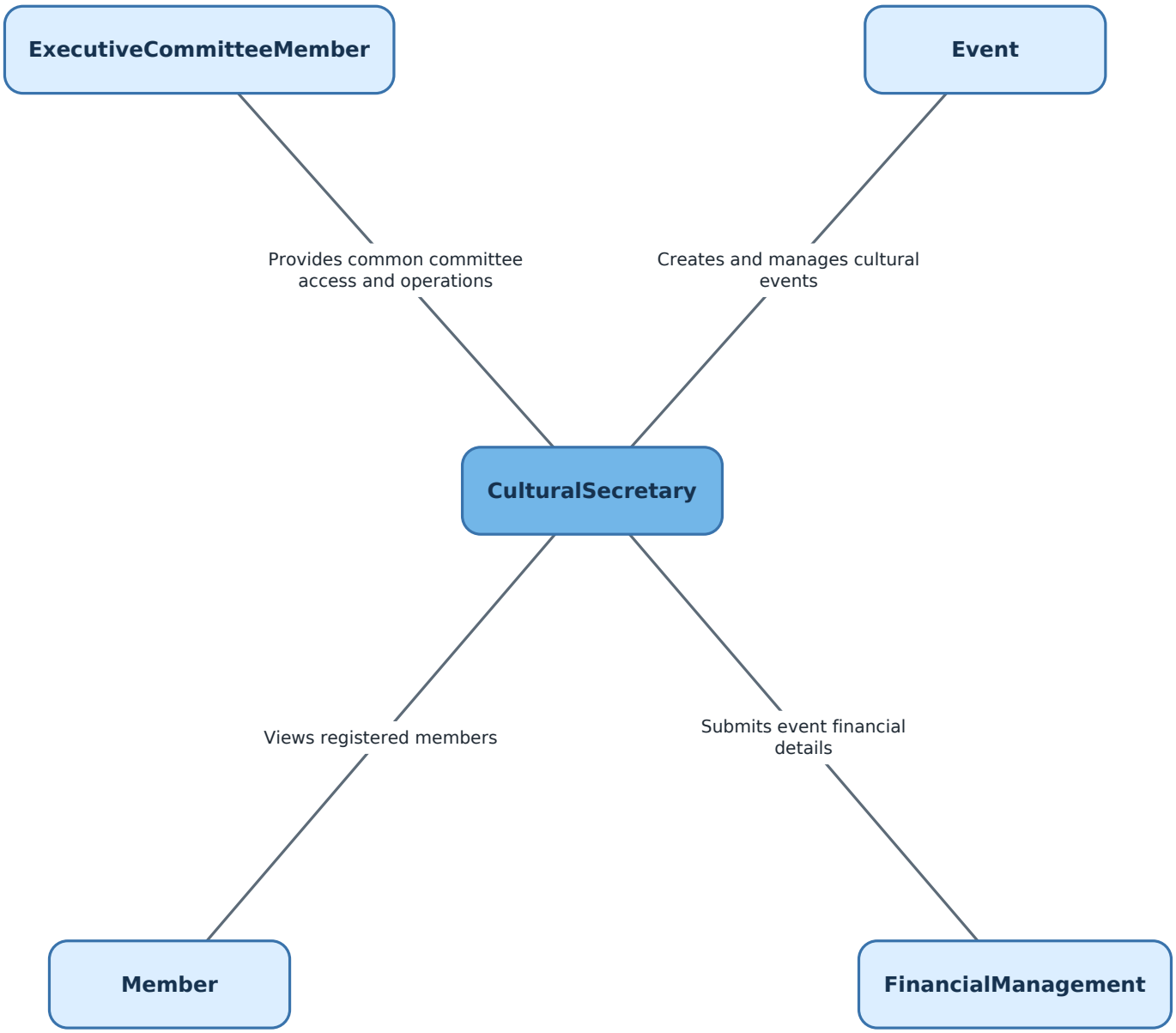


Fig crc.9: CulturalSecretary

Ref: Table 19

Name: FieldEquipmentSecretary

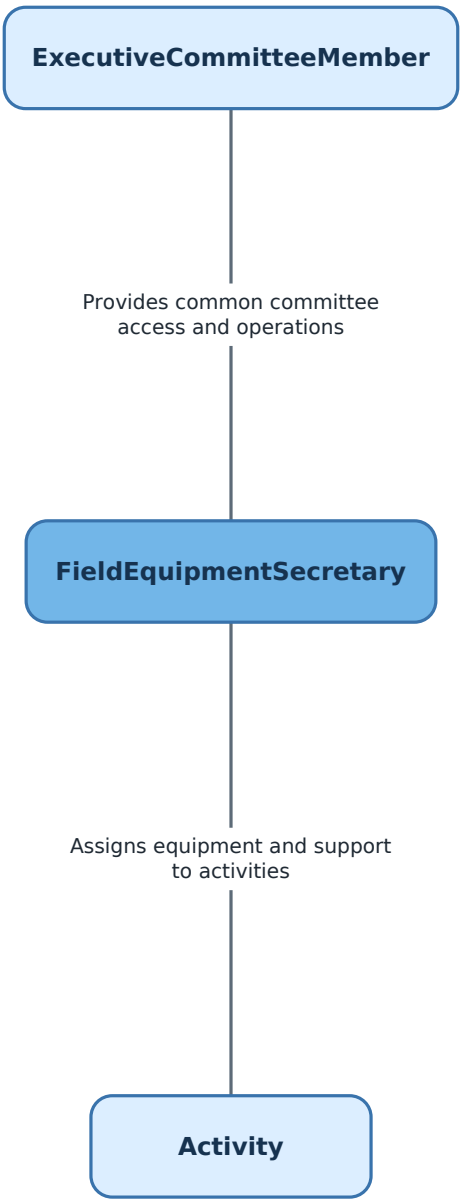


Fig crc.10: FieldEquipmentSecretary

Ref: Table 20

Name: Activity

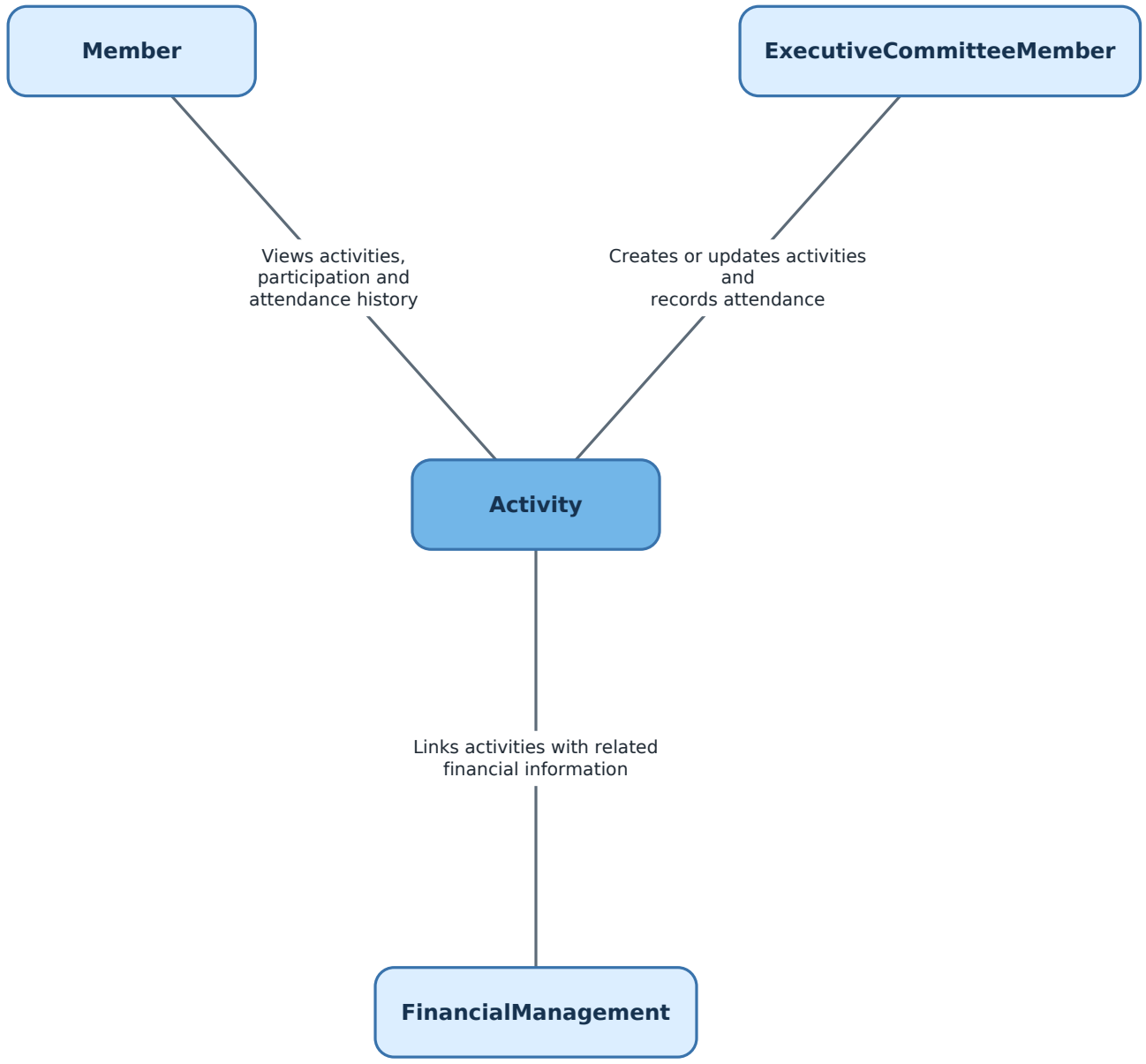


Fig crc.11: Activity

Ref: Table 21

Name: YogaSession

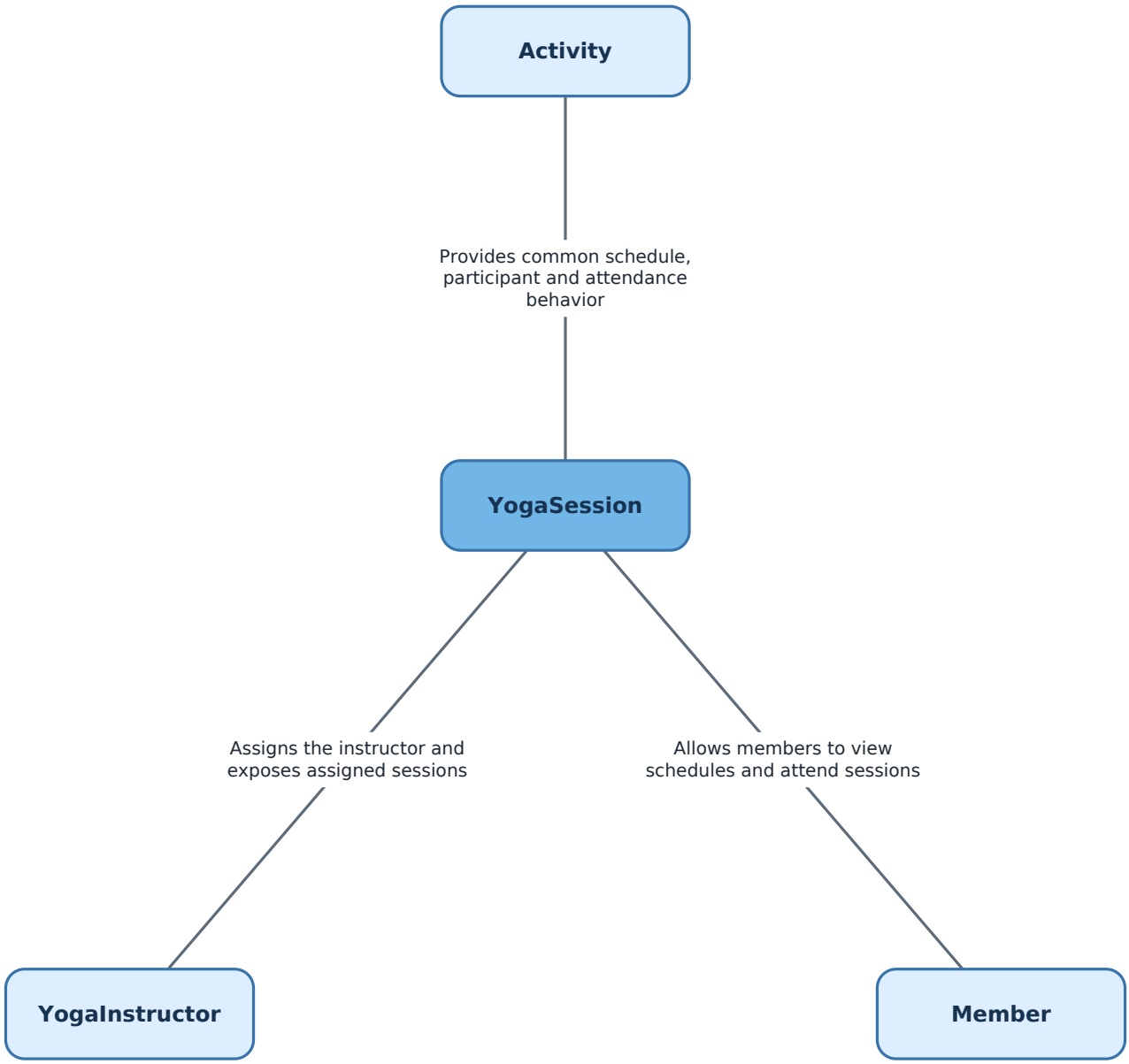


Fig crc.12: YogaSession

Ref: Table 23

Name: FinancialManagement

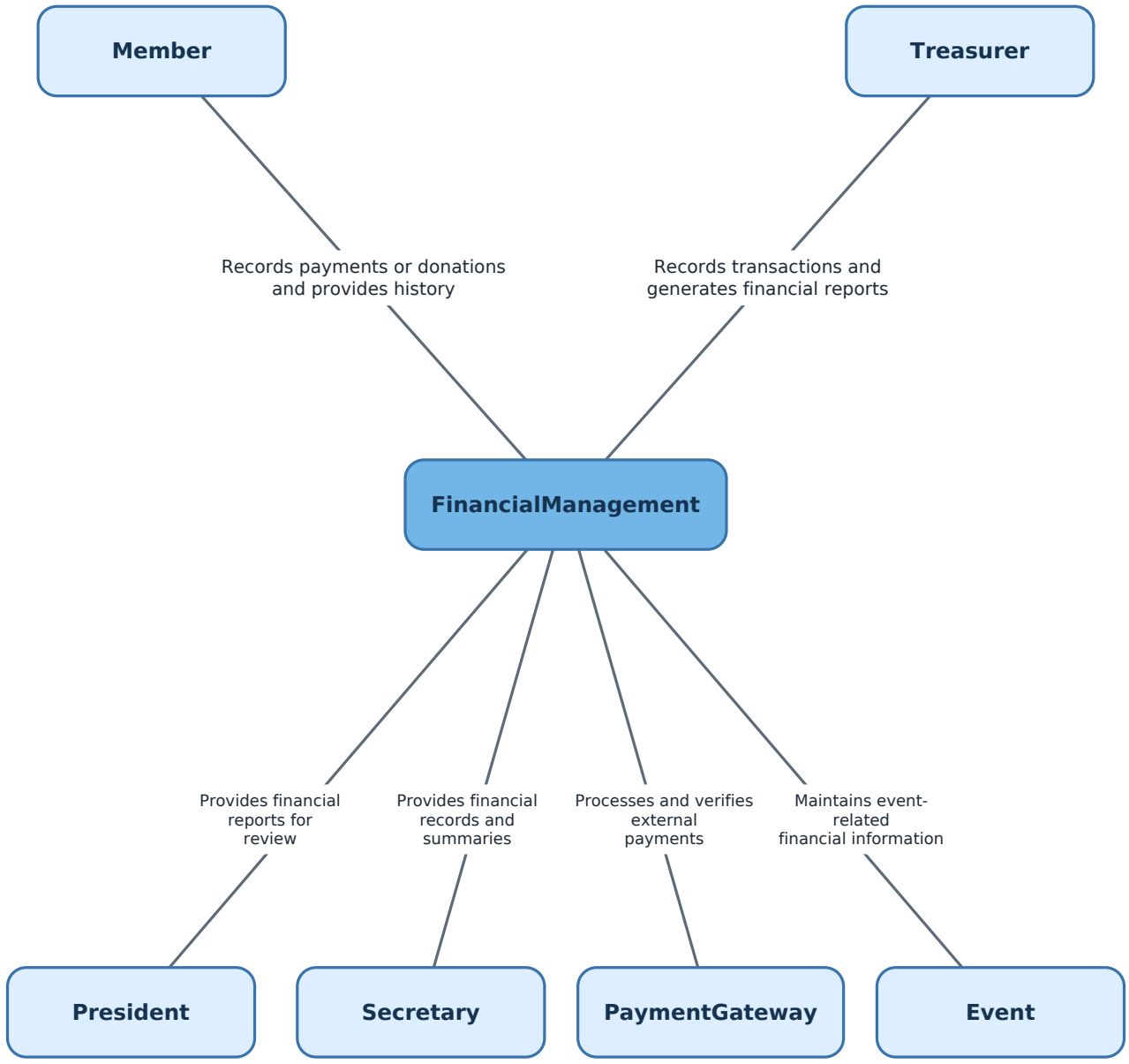


Fig crc.14: FinancialManagement

Ref: Table 24

Name: PaymentGateway

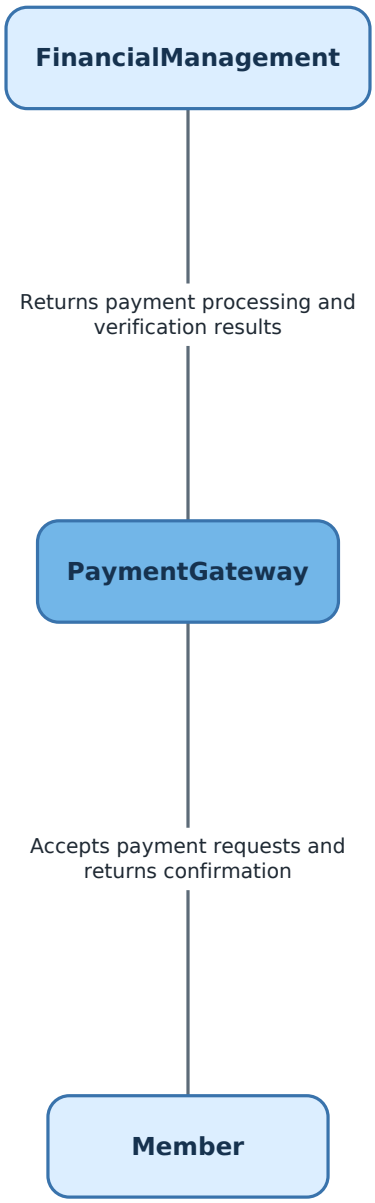


Fig crc.15: PaymentGateway

Ref: Table 25

Name: MediaPost

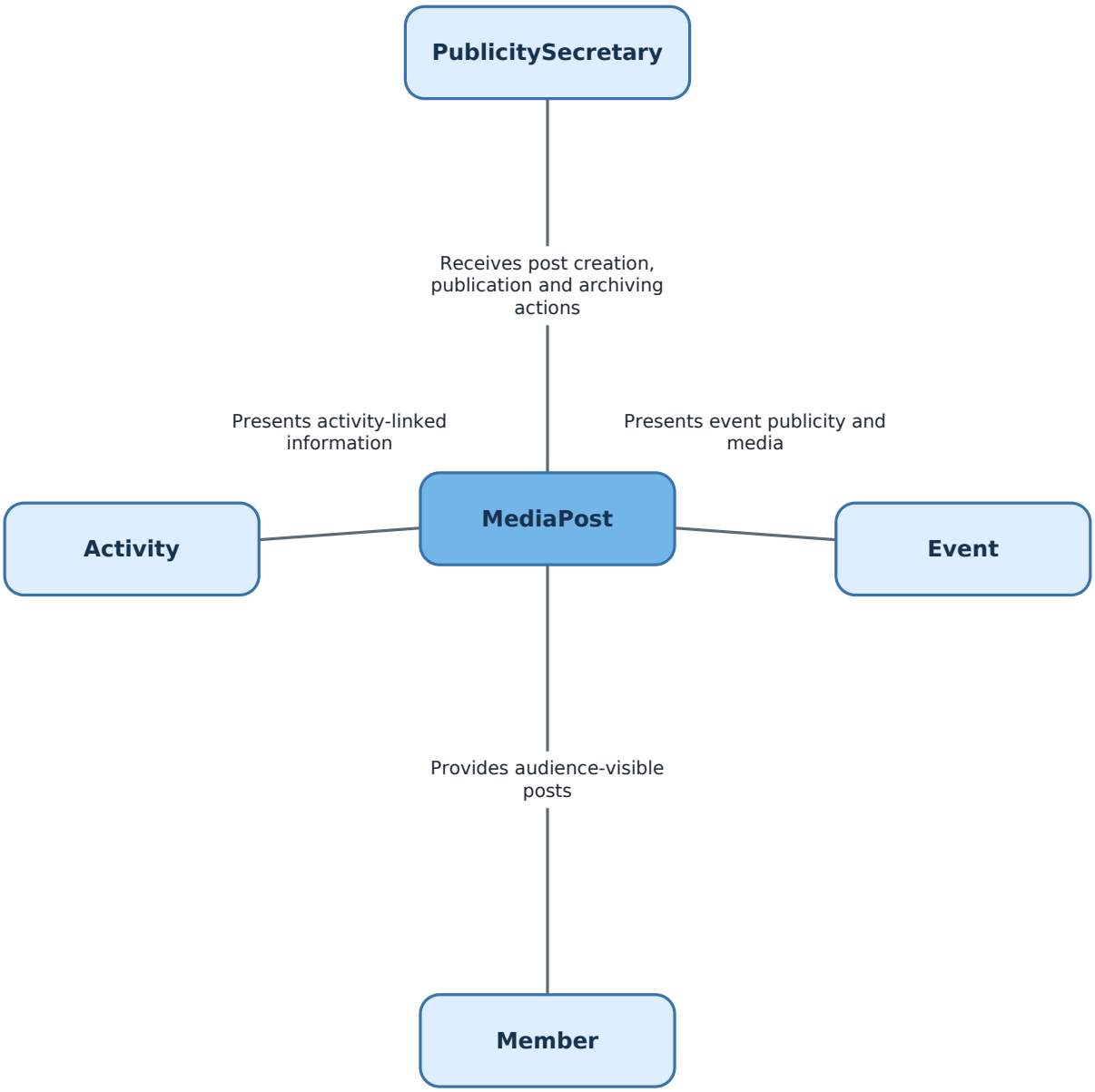


Fig crc.16: MediaPost

Ref: Table 26

Name: YogaContent

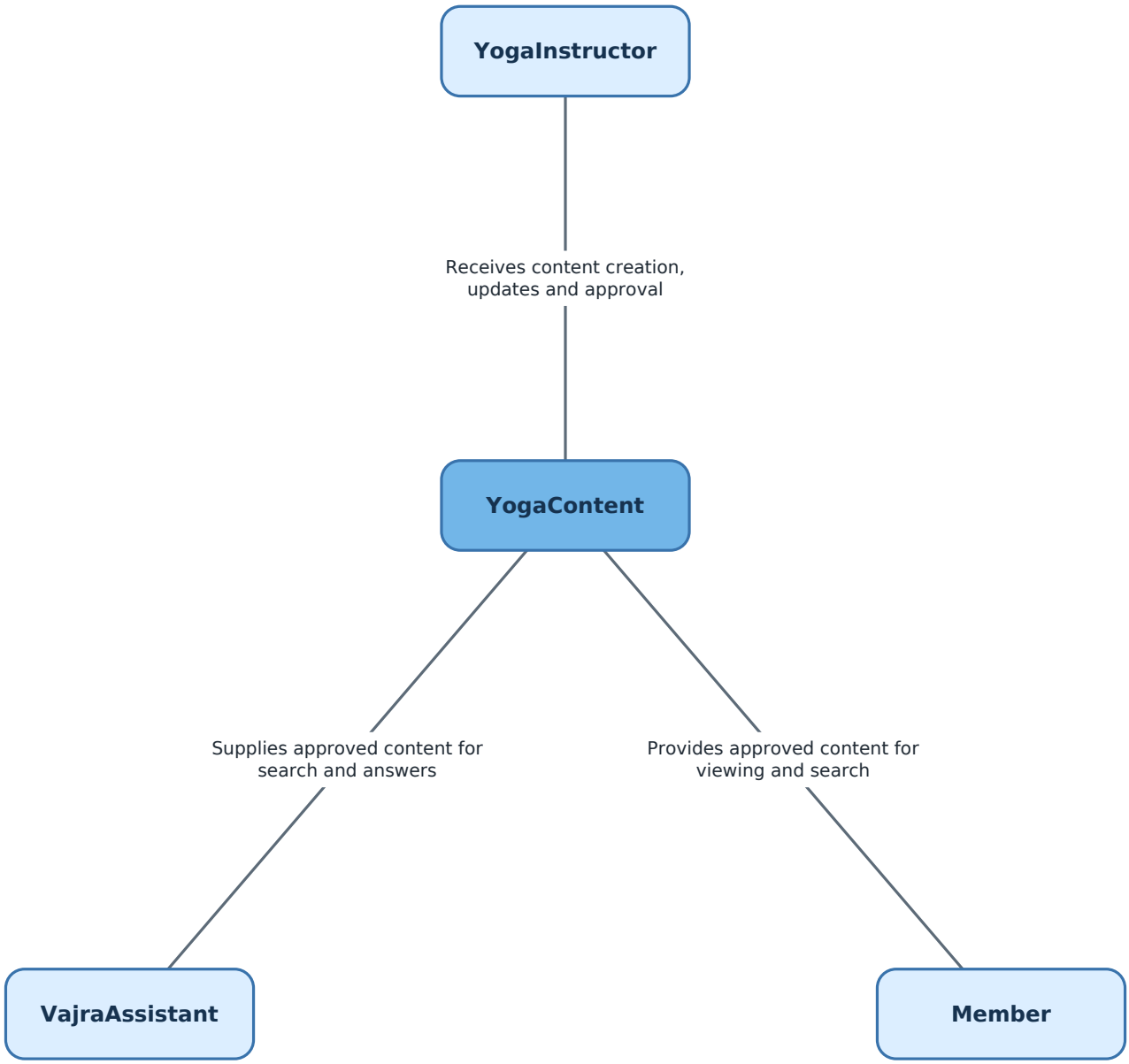


Fig crc.17: YogaContent

Ref: Table 27

Name: VajraAssistant

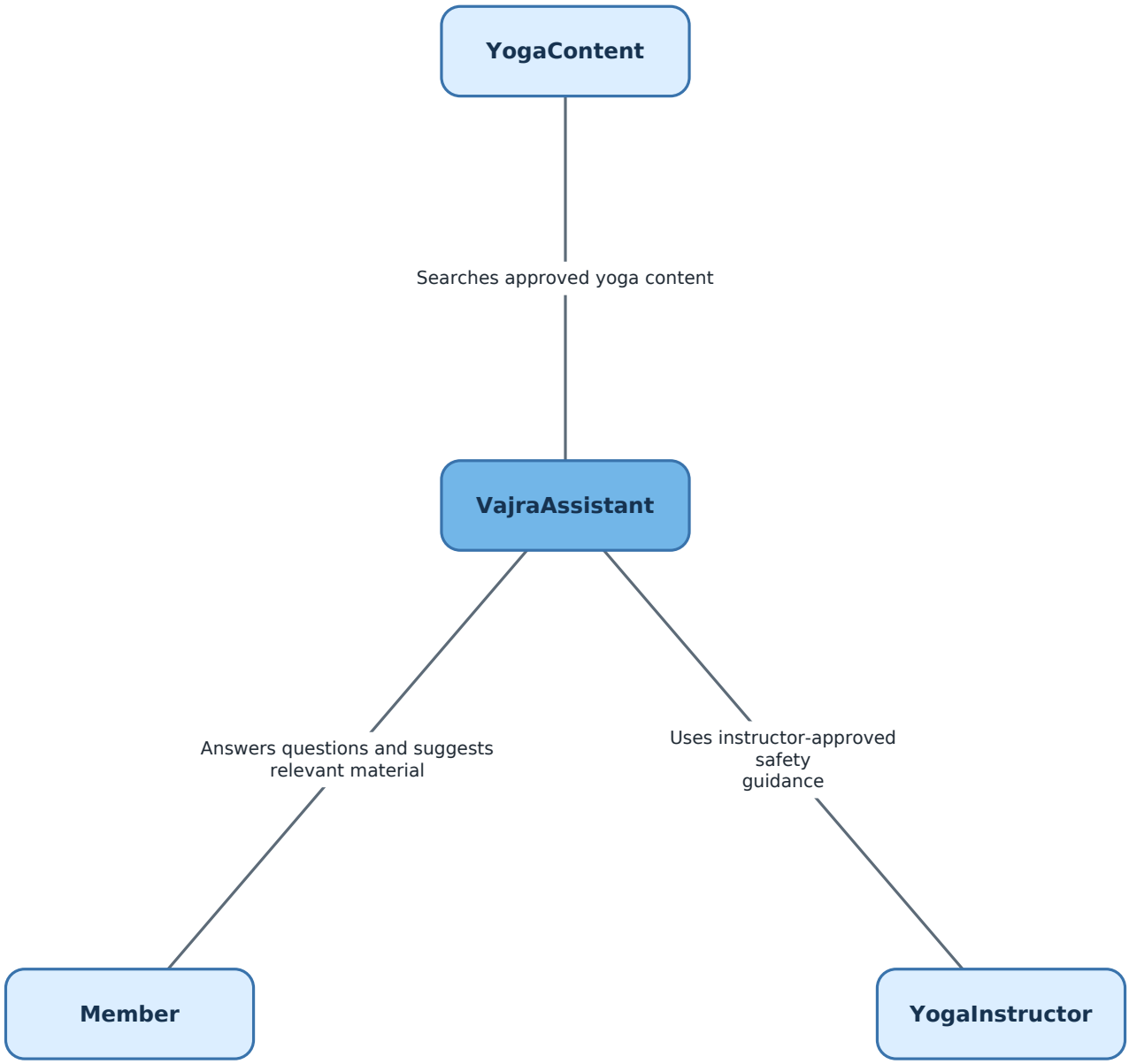


Fig crc.18: VajraAssistant